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Server, system, and management method

Abstract

A server includes a communication unit (first acquisition unit) that acquires information about a status of use of an EVSE by a user (first user) who uses the EVSE (power control device) in a normal state. Further, the server includes a processor (controller) that performs, based on the information about the status of use, control for lending the EVSE to a user (second user) different from the foregoing user in a time period in which the foregoing user does not use the EVSE.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
12046780	12/2023	Leighton	N/A	H01M 8/04089
2018/0154792	12/2017	Wada	N/A	B60L 53/14
2020/0101850	12/2019	Harty	N/A	B60L 55/00
2021/0380014	12/2020	Hishida	N/A	H02J 7/00
2022/0289067	12/2021	Adegbohun	N/A	H01M 10/6554
2023/0110777	12/2022	Neligan	320/109	B60L 53/30
2024/0185150	12/2023	Huang	N/A	G06Q 10/06

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
103794045	12/2013	CN	G08G 1/01
2018-093658	12/2017	JP	N/A
20180071150	12/2017	KR	G06Q 30/0254
WO-2015185009	12/2014	WO	G08G 1/0967

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This nonprovisional application is based on Japanese Patent Application No. 2022-125614 filed on Aug. 5, 2022 with the Japan Patent Office, the entire contents of which are hereby incorporated by reference.

BACKGROUND

Field

(2) The present disclosure relates to a server, a system, and a management method.

Description of the Background Art

(3) Japanese Patent Laying-Open No. 2018-093658 discloses a charging device provided at a user's home. By connecting a vehicle to the charging device via a charging cable, the vehicle is charged. When the user goes out, the vehicle is charged by connecting the vehicle to a charging device at a location to which the user goes.

SUMMARY

(4) In Japanese Patent Laying-Open No. 2018-093658, the charging device at home is not being used when the user goes out with the vehicle. Therefore, it has been desired to effectively utilize the charging device (power control device) that is not being used.

(5) The present disclosure has been made to solve the above-described problem and has an object to provide a server, a system, and a management method so as to effectively utilize a power control device that is not being used.

(6) A server according to a first aspect of the present disclosure is a server that manages at least one power control device capable of performing power control including at least one of power feeding from an electrically powered vehicle to a power system and charging from the power system to the electrically powered vehicle. The electrically powered vehicle is capable of performing the power control when the electrically powered vehicle is electrically connected to the power control device. The server includes: a first acquisition unit that acquires information about a status of use of the power control device by a first user who uses the power control device in a normal state; and a controller that performs, based on the information about the status of use, control for lending the power control device to a second user different from the first user in a time period in which the first user does not use the power control device.

(7) In the server according to the first aspect of the present disclosure, as described above, the control for lending the power control device to the second user different from the first user is performed in the time period in which the first user does not use the power control device. Thus, in the time period in which the power control device is not used by the first user, the power control device can be used by the second user. As a result, the power control device that is not being used can be effectively utilized.

(8) In some embodiments of the server according to the first aspect, the controller performs control of transmitting location information of the power control device to the second user. Thus, the second user can readily know the location of the power control device. As a result, the power control device that is not being used can be utilized more effectively.

(9) In some embodiments, the controller performs the control of transmitting the location information of the power control device to the second user who has requested to use the power control device. Thus, the power control device that is not being used can be effectively utilized while satisfying the need of the second user who desires to use the power control device.

(10) In some embodiments of the server according to the first aspect, the controller presents, on a bulletin board on the Internet, information of the power control device that is not being used by the first user. Thus, by using the Internet, the second user can readily acquire the information of the power control device that is not being used by the first user.

(11) In some embodiments, the at least one power control device includes a plurality of power control devices. When each of the plurality of power control devices is not being used by the first user, the controller presents, on the bulletin board, information of each of the plurality of power control devices. With such a configuration, by using the Internet, the second user can readily acquire the information of each of the plurality of power control devices that are not being each used by the first user.

(12) In some embodiments of the server that presents the information of each of the plurality of power control devices on the bulletin board, the controller controls a priority for each of the plurality of power control devices to be lent to the second user, by controlling an order of presentation of the plurality of power control devices on the bulletin board. With such a configuration, a power control device having a higher priority can be preferentially lent to the second user.

(13) In some embodiments, the controller controls the order of presentation to present, at a higher rank on the bulletin board, a power control device included in the plurality of power control devices and installed at a location at which a traffic amount is smaller. With such a configuration, a frequency of use of the power control device installed at the location at which a traffic amount is

smaller can be increased and a frequency of use of a power control device installed at a location at which a traffic amount is larger can be decreased. As a result, traffic congestion can be suppressed.

(14) In some embodiments of the server that controls the priority in accordance with the order of presentation, the electrically powered vehicle performs the power control in accordance with a supply/demand adjustment request for power in each of a plurality of regions, and the controller controls the order of presentation to present, at a higher rank on the bulletin board, a power control device included in the plurality of power control devices and located in a region in which the supply/demand adjustment request is larger. With such a configuration, a frequency of performing the power control can be increased in a region in which the supply/demand adjustment request for power is large. As a result, the supply/demand adjustment request for power can be readily satisfied.

(15) In some embodiments of the server that presents the information of each of the plurality of power control devices on the bulletin board, the plurality of power control devices include a first control device presented on the bulletin board when the first control device is not being used by the first user, and a second control device not presented on the bulletin board even when the second control device is not being used by the first user. When supply of the first control device is insufficient with respect to a demand for the plurality of power control devices, the controller performs control of requesting the first user, who uses the second control device in the normal state, to lend the second control device. With such a configuration, when the supply of the first control device is insufficient with respect to the demand for the power control devices, the second control device can be lent to the first user to whom the first control device cannot be lent.

(16) In some embodiments of the server that presents the information of the power control device on the bulletin board on the Internet, the controller presents, on the bulletin board, information of a size of a vehicle body that is able to be parked in a parking area in which the power control device is installed. With such a configuration, the second user can readily select the power control device to be used, based on the information about the size of the vehicle body.

(17) In some embodiments of the server according to the first aspect further includes a second acquisition unit that acquires information of each of an expected traveling route and an expected traveling time period of the electrically powered vehicle. The at least one power control device includes a plurality of power control devices. When each of the plurality of power control devices is not being used by the first user, the controller selects a power control device to be lent to the second user, based on at least one of the expected traveling route and the expected traveling time period. With such a configuration, an appropriate (optimal) power control device can be proposed to the second user based on at least one of the expected traveling route and the expected traveling time period of the second user. As a result, convenience for the second user can be increased.

(18) In some embodiments of the server according to the first aspect, the controller performs control of requesting the first user of the power control device in a region in which a frequency of lending the power control device to the second user is higher than a predetermined criterion, to permit the power control device to be lent also in a midnight period. With such a configuration, in a region in which a frequency of lending the power control device is relatively high, a period of time during which the power control device can be lent can be made long.

(19) A system according to a second aspect of the present disclosure includes: a power control device capable of performing power control including at least one of power feeding to a power system and charging from the power system; a server that manages the power control device; and an electrically powered vehicle capable of performing the power control when the electrically powered vehicle is electrically connected to the power control device. The server acquires information about a status of use of the power control device by a first user who uses the power control device in a normal state, and performs, based on the information about the status of use, control for lending the power control device to a second user different from the first user in a time period in which the first user does not use the power control device.

(20) In the system according to the second aspect of the present disclosure, as described above, the control for lending the power control device to the second user different from the first user is performed in the time period in which the first user does not use the power control device. Thus, in the time period in which the power control device is not used by the first user, the power control device can be used by the second user. As a result, the system can be provided to effectively utilize the power control device that is not being used.

(21) A management method according to a third aspect of the present disclosure is a management method using a server that manages a power control device capable of performing power control including at least one of power feeding from an electrically powered vehicle to a power system and charging from the power system to the electrically powered vehicle. The electrically powered vehicle is capable of performing the power control when the electrically powered vehicle is electrically connected to the power control device. The management method includes: acquiring information about a status of use of the power control device by a first user who uses the power control device in a normal state; and performing, based on the information about the status of use, control for lending the power control device to a second user different from the first user in a time period in which the first user does not use the power control device.

(22) In the management method according to the third aspect of the present disclosure, as described above, the control for lending the power control device to the second user different from the first user is performed in the time period in which the first user does not use the power control device. Thus, in the time period in which the power control device is not used by the first user, the power control device can be used by the second user. As a result, the management method can be provided to effectively utilize the power control device that is not being used.

(23) The foregoing and other objects, features, aspects and advantages of the present disclosure will become more apparent from the following detailed description of the present disclosure when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a diagram showing a configuration of a system according to a first embodiment.
- (2) FIG. 2 is a diagram showing a bulletin board on the Internet according to the first embodiment.
- (3) FIG. 3 is a first diagram showing a sequence in the system according to the first embodiment.
- (4) FIG. 4 is a second diagram showing a sequence (route A) in the system according to the first embodiment.
- (5) FIG. 5 is a third diagram showing a sequence in the system according to the first embodiment.
- (6) FIG. 6 is a diagram showing display of a navigation system of an electrically powered vehicle according to the first embodiment.
- (7) FIG. 7 is a diagram showing a configuration of a system according to a second embodiment.
- (8) FIG. 8 is a diagram showing a sequence in the system according to the second embodiment.

DESCRIPTION OF THE EMBODIMENTS

(9) Hereinafter, embodiments of the present disclosure will be described in detail with reference to figures. In the figures, the same or corresponding portions are denoted by the same reference characters and will not be described repeatedly.

First Embodiment

(10) FIG. 1 is a diagram showing a configuration of a system **1** according to a first embodiment. System **1** includes a power system PG, a server **100**, an electrically powered vehicle **10**, and a plurality of EVSEs (Electric Vehicle Supply Equipment) **20**. Server **100** manages the plurality of EVSEs **20**. It should be noted that each of EVSEs **20** is an example of the “power control device” of the present disclosure.

(11) EVSE **20** can feed power from electrically powered vehicle **10** to power system PG (external power feeding) and can charge power from power system PG to electrically powered vehicle **10** (external charging). Hereinafter, the external power feeding and the external charging may be collectively referred to as power control. Electrically powered vehicle **10** can perform the power control when electrically powered vehicle **10** is electrically connected to EVSE **20**.

(12) EVSEs **20** include an EVSE **20A**, an EVSE **20B**, an EVSE **20C**, and an EVSE **20D**. EVSE **20A**, EVSE **20B**, EVSE **20C**, and EVSE **20D** are provided at (provided beside) a house **21A**, a house **21B**, a house **21C**, and a house **21D**, respectively. It should be noted that each of EVSE **20A**, EVSE **20B**, and EVSE **20C** is an example of the “first control device” of the present disclosure. EVSE **20D** is an example of the “second control device” of the present disclosure.

(13) It should be noted that each of EVSEs **20** is used by a user **22** in a normal state. Specifically, EVSE **20A**, EVSE **20B**, EVSE **20C**, and EVSE **20D** are respectively used by a user **22A**, a user **22B**, a user **22C**, and a user **22D** in the normal state. For example, users **22A** to **22D** are residents of houses **21A** to **21D**, respectively. It should be noted that the normal state means, for example, that user **22A** stays in house **21A** (that is, a vehicle of user **22A** is parked at house **21A**).

(14) Examples of electrically powered vehicle **10** include a PHEV (Plug-in Hybrid Electric Vehicle), a BEV (Battery Electric Vehicle), and an FCEV (Fuel Cell Electric Vehicle). It should be noted that electrically powered vehicle **10** is used by a user **11**. User **11** does not use EVSE **20** in the normal state.

(15) Power system PG is a power grid constituted of a power plant and a power transmission/distribution facility (not shown). In the first embodiment, a power company serves as both a power generation entity and a power transmission/distribution entity. The power company corresponds to a general power transmission/distribution entity, and maintains and manages power system PG. The power company corresponds to an administrator of power system PG.

(16) Further, server **100** is configured to manage: information of a registered electrically powered vehicle **10** (hereinafter also referred to as “vehicle information”); information of a registered user (hereinafter also referred to as “user information”), and information of a registered EVSE **20** (hereinafter also referred to as “EVSE information”). The user information, the vehicle information, and the EVSE information are distinguished by identification information (ID) and are stored in a memory **102** of server **100**.

(17) A user ID is identification information for identifying a user, and also functions as information (terminal ID) for identifying a mobile terminal (not shown) carried by the user. Server **100** is configured to store pieces of information received from mobile terminals in a distinguished manner for respective users ID. The user information includes a communication address of the mobile terminal carried by the user and a vehicle ID of electrically powered vehicle **10** belonging to the user.

(18) The vehicle ID is identification information for identifying electrically powered vehicle **10**. The vehicle ID may be a license plate or VIN (Vehicle Identification Number). The vehicle information includes a schedule of traveling of electrically powered vehicle **10**.

(19) An EVSE-ID is identification information for identifying EVSE **20**. The EVSE information includes a communication address of EVSE **20** and a state of electrically powered vehicle **10** connected to EVSE **20**. Further, the EVSE information includes information indicating a combination of electrically powered vehicle **10** and EVSE **20** connected to each other (for example, a combination of the EVSE-ID and the vehicle ID).

(20) Server **100** includes a processor **101**, memory **102**, and a communication unit **103**. It should be noted that processor **101** is an example of the “controller” of the present disclosure. Communication unit **103** is an example of each of the “first acquisition unit” and the “second acquisition unit” of the present disclosure.

(21) Memory **102** stores: a program to be executed by processor **101**; and information (for example, a map, a mathematical formula, and various types of parameters) to be used by the

program. Communication unit **103** includes various types of communication I/Fs. Processor **101** controls communication unit **103**.

(22) Communication unit **103** can communicate with a DCM (Data Communication Module) (not shown) of electrically powered vehicle **10** and each of EVSEs **20A** to **20D**. Further, server **100** can use (access) Internet **200** through communication unit **103**. Communication unit **103** acquires information of each of an expected traveling route and an expected traveling time period of electrically powered vehicle **10**. It should be noted that communication unit **103** acquires each of the above-described pieces of information through communication from the DCM (not shown) of electrically powered vehicle **10** and a terminal (not shown) or the like possessed by user **11**.

(23) Communication unit **103** acquires information about a status of use of EVSE **20** by user **22** who uses EVSE **20** in the normal state. For example, communication unit **103** detects a use state (power state) of EVSE **20** to acquire the information about the status of use of EVSE **20**. The status of use includes information about a use state at the present time and information about expected use in future.

(24) Here, in a conventional system, an EVSE may not be effectively utilized due to presence of a time period in which the EVSE is not being used (not expected to be used).

(25) In view of this, in the first embodiment, processor **101** performs, based on the information about the status of use of EVSE **20**, control for lending EVSE **20** to user **11** different from user **22** in a time period in which user **22** does not use EVSE **20**.

(26) Specifically, as shown in FIG. 2, processor **101** presents, on a bulletin board **201** on Internet **200**, information of EVSE **20** that is not being used by user **22**. FIG. 2 shows an example in which each of EVSE **20A** and EVSE **20B** is not being used by user **22** (**22A**, **22B**) and can be lent at the present time. Further, bulletin board **201** shown in FIG. 2 presents information of EVSE **20C** that cannot be lent because it is currently being used by user **22C**.

(27) It should be noted that EVSE **20D** is an EVSE that is not presented on bulletin board **201** even when EVSE **20D** is not currently being used by user **22D**. EVSE **20D** is set not to be presented on bulletin board **201** (by contract or the like) even when EVSE **20D** is not currently being used. It should be noted that there may be a plurality of EVSEs **20** (**20D**) not presented on bulletin board **201** even when they are not currently being used.

(28) Various pieces of information about EVSEs **20** that can be lent are presented on bulletin board **201**. Specifically, pieces of location information (addresses) of EVSEs **20** that can be lent are presented on bulletin board **201**.

(29) Further, bulletin board **201** presents information about the size of a vehicle body that can be parked in a parking area in which EVSE **20** that can be lent is installed. Bulletin board **201** shown in FIG. 2 indicates that there is no restriction on the size of the vehicle body that can be parked in the parking area in which EVSE **20A** is installed. On the other hand, bulletin board **201** shown in FIG. 2 presents that there is a restriction on the size (total length and vehicle width) of the vehicle body in the parking area in which EVSE **20B** is installed.

(30) Further, bulletin board **201** presents information about specifications (a specification regarding charging/discharging, a specification regarding communication, and the like) of EVSE **20** that can be lent. Bulletin board **201** shown in FIG. 2 presents that EVSEs **20A** and **20B** correspond to normal charging and rapid charging, respectively.

(31) Further, bulletin board **201** presents information about an available time of EVSE **20** that can be lent. Bulletin board **201** shown in FIG. 2 presents that EVSE **20A** can be used in a period from the present time to 15 o'clock and EVSE **20B** can be used in a period from the present time to 20 o'clock.

(32) Further, bulletin board **201** presents information about EVSE **20** (**20C**) that cannot be lent. Bulletin board **201** shown in FIG. 2 presents the location of EVSE **20C** the size of the vehicle body that can be parked, and the information about the specifications. Further, bulletin board **201** presents information about a time when EVSE **20C**, which cannot be currently lent, becomes

available. Specifically, bulletin board **201** shown in FIG. 2 presents that EVSE **20C** will become available from 17 o'clock.

(33) Further, processor **101** controls an order of presentation of the plurality of EVSEs **20** on bulletin board **201**. Specifically, when there are a plurality of EVSEs **20** that can be lent, processor **101** controls the order of presentation of the plurality of EVSEs **20** that can be lent. It should be noted that when there are a plurality of EVSEs **20** that cannot be lent, processor **101** may control an order of presentation of EVSEs **20** that cannot be lent.

(34) Specifically, processor **101** controls the order of presentation to present, at a higher rank on bulletin board **201**, an EVSE **20** included in the plurality of EVSEs **20** (that can be lent) and installed at a location at which a traffic amount is smaller (for example, a location away from a principal road). On bulletin board **201** shown in FIG. 2, EVSE **20A** is presented at a higher rank than EVSE **20B** because a traffic amount at the location at which EVSE **20A** is installed is smaller than a traffic amount at the location at which EVSE **20B** is installed. With this, EVSE **20A** is promoted to be lent as compared with EVSE **20B**. Thus, processor **101** can control a priority for each of the plurality of EVSEs **20** to be lent to user **11**.

(35) Here, memory **102** (see FIG. 1) stores information about a magnitude of a traffic amount at the location at which each of EVSEs **20** is installed. Processor **101** controls the order of presentation of EVSEs **20** based on the information about the magnitude of the traffic amount stored in memory **102**. It should be noted that server **100** may acquire, in real time (or whenever a predetermined time elapses) through communication unit **103**, the information of the traffic amount at the location at which each EVSE **20** is installed.

(36) Further, on bulletin board **201**, user **11** performs an operation of selecting any one of EVSEs **20** that can be lent and that are presented on bulletin board **201**, thereby permitting user **11** to lend EVSE **20**. It should be noted that in addition to use from the present time, use from a predetermined time in future can be reserved. Further, user **11** of electrically powered vehicle **10** can desire (apply) random lending of EVSE **20** without specifying a type of EVSE **20**.

(37) (Management Method)

(38) Next, a management method using server **100** will be described with reference to sequence diagrams of FIGS. 3 to 5. It should be noted that in the sequence shown in each of FIGS. 3 to 5, it is assumed that there is at least one EVSE **20** that can be lent.

(39) First, in a step S1, each of the plurality of EVSEs **20** transmits the information about its status of use to server **100** (communication unit **103**). Next, in a step S2, server **100** (communication unit **103**) receives the information about the status of use transmitted from EVSE **20** in step S1.

(40) In a step S3, server **100** (processor **101**) determines whether or not the number of EVSEs **20** that can be lent is insufficient, based on the information acquired in step S2. When it is determined that the number of EVSEs **20** that can be lent is insufficient (Yes in S3), the process proceeds to a step S4. When it is determined that the number of EVSEs **20** that can be lent is sufficient (No in S3), the process proceeds to a step S5. It should be noted that the expression “the number of EVSEs **20** is insufficient” means that the number of EVSEs **20** that can be lent among EVSEs **20A** to **20C** is insufficient with respect to a demand for EVSEs **20**.

(41) A method of determination in step S3 will be specifically described. For example, when there is no EVSE **20** that can be lent, server **100** (processor **101**) determines that the number of EVSEs **20** that can be lent is insufficient. Further, when processor **101**, which is configured to detect the number of users **11** who are browsing bulletin board **201**, detects that the number of detected users **11** is more than the number of EVSEs **20** that can be lent, it may be determined that the number of EVSEs **20** that can be lent is insufficient. It should be noted that when the number of users **11** who are viewing bulletin board **201** is equal to or more than a predetermined number, it may be determined that the number of EVSEs **20** is insufficient. Further, when the number of users who desire to lend unspecific EVSEs **20** (who allow server **100** to randomly select EVSEs **20** to be lent) is more than the number of EVSEs **20** that can be lent, it may be determined that the number of

EVSEs **20** is insufficient.

(42) In a step **S4**, processor **101** performs control of requesting user **22D** who uses EVSE **20D** in the normal state to lend (open up) EVSE **20D**. When the request is accepted by user **22D**, the information of EVSE **20D** is presented on bulletin board **201**. After step **S4**, the process proceeds to a step **S5**. It should be noted that when there are a plurality of users **22D** (EVSEs **20D**), a request for lending is made to a user **22D** different from the previous user in step **S4** performed for the second or subsequent time.

(43) In step **S5**, processor **101** sets a higher priority on an EVSE **20** included in EVSEs **20** that can be lent and installed at a location at which a traffic amount is smaller. Next, in a step **S6**, processor **101** controls the order of presentation (see FIG. 2) to present, at a higher rank on bulletin board **201**, EVSE **20** having the higher priority set in step **S5**.

(44) In a step **S7**, it is assumed that user **11** of electrically powered vehicle **10** has applied for use of a specific EVSE **20**. In the description below, it is assumed that user **11** has applied for use of EVSE **20A**.

(45) In a step **S8**, processor **101** transmits, to electrically powered vehicle **10** (or the terminal of user **11**) through communication unit **103**, detailed location information of EVSE **20A** for which the application for use has been made in step **S7**.

(46) In a step **S9**, the location information transmitted in step **S8** is displayed on a navigation system **12** (see FIG. 6) of electrically powered vehicle **10**. The location information displayed on navigation system **12** includes more detailed information than the location information (see FIG. 2) displayed on the bulletin board. For example, navigation system **12** displays address information **12a** of the location at which EVSE **20** is installed. Further, a map **12b** is displayed on navigation system **12** to indicate the location at which EVSE **20** is installed.

(47) Further, navigation system **12** displays a message for confirming whether to set, to a destination, a location that is based on the displayed location information. When a button **12c** representing “YES” is selected, the location that is based on the location information is set to the destination. When a button **12d** representing “NO” is selected, it is not set to the destination. It should be noted that the location that is based on the location information may be automatically set to the destination without performing the above confirmation.

(48) In a step **S10**, processor **101** determines whether or not the version of a power control program of electrically powered vehicle **10** coincides with the version of a power control program of EVSE **20A** (whether or not there is connection compatibility). Here, the program of EVSE **20** is basically the latest version. Hence, in step **S10**, processor **101** determines whether or not the version of the power control program of electrically powered vehicle **10** is older than the version of the power control program of EVSE **20A**. When the versions are different from each other (Yes in **S10**), the process proceeds to a step **S11**. When the versions coincide with each other (No in **S10**), the process of server **100** is ended.

(49) In step **S11**, processor **101** transmits the same version of the program as the version of the power control program of EVSE **20A** to electrically powered vehicle **10** through communication unit **103**. In a step **S12**, the program transmitted in step **S11** is installed into electrically powered vehicle **10**. As a result, the power control program of electrically powered vehicle **10** is updated.

(50) Next, referring to FIG. 4, the following describes a sequence of a route A branching after step **S4** in FIG. 3.

(51) First, in a step **S21**, it is assumed that user **11** of electrically powered vehicle **10** has applied for use (lending) of an EVSE **20** without specifying EVSE **20**. In other words, user **11** desires random selection of an EVSE **20** to be lent.

(52) Next, in a step **S22**, server **100** (processor **101**) determines whether or not there are a plurality of EVSEs **20** that can be lent. When there are a plurality of EVSEs **20** that can be lent (Yes in **S22**), the process proceeds to a step **S23**. When there are not a plurality of EVSEs **20** that can be lent (i.e., when there is only one EVSE **20** that can be lent) (No in **S22**), the process of route A is ended.

(53) Next, in step S23, server **100** (processor **101**) acquires information about an expected traveling route and an expected traveling time period of electrically powered vehicle **10** (having applied for use of unspecific EVSE **20**) through communication unit **103**. It should be noted that the process of step S23 may be completed before this timing.

(54) Next, in a step S24, server **100** (processor **101**) selects an EVSE **20** suitable for the expected traveling route and expected traveling time period acquired in step S23. Specifically, an EVSE **20** provided along the expected traveling route, an EVSE **20** that can be lent in the expected traveling time period, or the like is selected. On this occasion, a plurality of EVSEs **20** may be selected. Then, the process proceeds to step S8 (see FIG. 3).

(55) Next, a sequence showing another control by server **100** will be described with reference to FIG. 5. It should be noted that the control shown in FIG. 5 may be performed at any timing.

(56) First, in a step S31, server **100** (processor **101**) determines whether or not there is a region in which a frequency of lending EVSE **20** is higher than a predetermined criterion. For example, when there is a region in which EVSE **20** is lent a predetermined number of times (for example, ten times) or more in a recent predetermined period (for example, one week), processor **101** determines that the region is a region in which the frequency is higher than the predetermined criterion. When it is determined that there is a region in which the frequency is higher than the predetermined criterion (Yes in S31), the process proceeds to a step S32. When it is determined that there is no region in which the frequency is higher than the predetermined criterion (No in S31), the process of step S31 is repeated.

(57) Next, in step S32, processor **101** performs control of requesting user **22** of EVSE **20** in the region in which the frequency of lending EVSE **20** is determined to be higher than the predetermined criterion in step S31, to lend EVSE **20** also in a midnight period. It should be noted that the request may be made to user **22** of EVSE **20** for which the frequency of lending is determined to be higher than the predetermined criterion without taking the frequency of lending in each region into consideration.

(58) As described above, in the first embodiment, processor **101** performs control for lending EVSE **20** to user **11** in a time period in which user **22** does not use EVSE **20**. Thus, it is possible to suppress presence of a period of time during which EVSE **20** is not being used. As a result, EVSE **20** can be effectively utilized.

(59) Further, in the first embodiment, processor **101** performs the control of transmitting the location information of EVSE **20** to user **11**. Thus, user **11** can readily use EVSE **20**, with the result that EVSE **20** can be utilized more effectively.

Second Embodiment

(60) Next, a second embodiment of the present disclosure will be described with reference to FIGS. 7 and 8. In the second embodiment, unlike the first embodiment in which the priority on each of EVSEs **20** to be lent is determined based on the magnitude of the traffic amount, the priority is determined based on a magnitude of a supply/demand adjustment request for power. It should be noted that the same configurations as those in the first embodiment are denoted by the same reference characters and will not be described repeatedly.

(61) FIG. 7 is a diagram showing a configuration of a system **2** according to a second embodiment. System **2** includes a power system PG, a server **300**, an electrically powered vehicle **10**, a plurality of EVSEs **20**, and a system management server **400**. It should be noted that in the second embodiment, it is assumed that the plurality of EVSEs **20** are provided in different regions.

(62) System management server **400** manages supply/demand of power system PG (power grid). System management server **400** belongs to a power company. Based on generation and consumption of power by each power adjustment resource managed by system management server **400**, system management server **400** transmits, to server **300**, a request (supply/demand adjustment request) for adjusting a power demand amount of power system PG. Specifically, when power to be generated by the power adjustment resource is expected to be larger (or when power generated by

the power adjustment resource is currently larger) than that in the normal state, system management server **400** transmits, to server **300**, a request for increasing the power demand amount to be larger than that in the normal state. Further, when power to be consumed by the power adjustment resource is expected to be larger (or when power consumed by the power adjustment resource is currently larger) than that in the normal state, system management server **400** transmits, to server **300**, a request for decreasing the power demand amount to be smaller than that in the normal state.

(63) Server **300** is a server managed by the aggregator. The aggregator is an electricity business entity that aggregates a plurality of power adjustment resources in regions and predetermined facilities to provide an energy management service.

(64) Server **300** requests electrically powered vehicle **10** to perform power control as one means for increasing or decreasing the power demand amount of power system PG. Server **300** transmits, to electrically powered vehicle **10** or a mobile terminal (not shown) possessed by the user of electrically powered vehicle **10**, a request signal for making the above-described request to electrically powered vehicle **10**. Electrically powered vehicle **10** performs the power control in accordance with a supply/demand adjustment request for power in each of a plurality of regions.

(65) Server **300** includes a processor **301**, a memory **302**, and a communication unit **303**. It should be noted that processor **301** is an example of the “controller” of the present disclosure.

Communication unit **303** is an example of each of the “first acquisition unit” and the “second acquisition unit” of the present disclosure.

(66) (Management Method)

(67) Next, a management method using server **300** will be described with reference to a sequence diagram of FIG. **8**. In the sequence diagram of FIG. **8**, a process of step **S15** is performed instead of the process of step **S5** in the sequence diagram of the first embodiment (see FIG. **3**). It should be noted that processes other than step **S15** are the same as those in the first embodiment.

(68) In step **S15**, server **300** (processor **301**) controls the order of presentation so as to present, at a higher rank on bulletin board **201**, an EVSE **20** included in the plurality of EVSEs **20** and installed in a region in which the supply/demand adjustment request for power is larger. Whenever a predetermined time elapses, server **300** acquires, through communication unit **303**, information (such as the magnitude of the request) about the supply/demand adjustment request in the region in which each EVSE **20** is provided. Then, processor **301** controls the order of presentation of EVSEs **20** on bulletin board **201** based on the acquired information about the supply/demand adjustment request.

(69) It should be noted that configurations and controls other than those described above are the same as those in the first embodiment, and therefore will not be described repeatedly.

(70) As described above, in the second embodiment, processor **301** controls the order of presentation to present, at a higher rank on bulletin board **201**, an EVSE **20** included in the plurality of EVSEs **20** and located in the region in which the supply/demand adjustment request for power is larger. Thus, the power control can be more promoted in the region in which the supply/demand adjustment request for power is large.

(71) In each of the above-described first and second embodiments, it has been illustratively indicated that when an application has been made to use a specific EVSE location information of EVSE **20** is transmitted to user **11** having made the application; however, the present disclosure is not limited thereto. For example, the location information may be presented on bulletin board **201** of Internet **200**. In this case, control may be performed such that only the user having made the application can view the location information.

(72) In each of the first and second embodiments, it has been illustratively indicated that information of EVSE **20** that is not being used by user **22** is presented on bulletin board **201**; however, the present disclosure is not limited thereto. For example, information of EVSE **20** that is not being used by user **22** may be transmitted to user **11** of electrically powered vehicle **10** without

presenting it on bulletin board **201**.

(73) In each of the first and second embodiments, it has been illustratively indicated that the program is transmitted to electrically powered vehicle **10** to update the program of electrically powered vehicle **10**; however, the present disclosure is not limited thereto. A program may be sent to EVSE **20** to update the program of EVSE **20**.

(74) In each of the above-described first and second embodiments, it has been illustratively indicated that EVSE **20** having a high priority is presented at a higher rank on bulletin board **201**; however, the present disclosure is not limited thereto. For example, information of EVSE **20** having a higher priority may be displayed in a larger size on bulletin board **201**.

(75) Although the present disclosure has been described and illustrated in detail, it is clearly understood that the same is by way of illustration and example only and is not to be taken by way of limitation, the scope of the present disclosure being interpreted by the terms of the appended claims.

Claims

1. A server comprising: a processor that: acquires information about a status of use of a power control device by a first user who uses the power control device in a normal state, performs, based on the information about the status of use, control for lending the power control device to a second user different from the first user in a time period in which the first user does not use the power control device, controls the power control device that is configured to perform power control including feeding power from an electrically powered vehicle to a power system or charging power from the power system to the electrically powered vehicle, the electrically powered vehicle being electrically connected to the processor, and acquires, in response to lapse of a predetermined time, information in real-time of a traffic amount at a location at which the power control device is installed.
2. The server according to claim 1, wherein the processor performs control of transmitting location information of the power control device to the second user.
3. The server according to claim 2, wherein the processor performs the control of transmitting the location information of the power control device to the second user who has requested to use the power control device.
4. The server according to claim 1, wherein the processor presents, on a bulletin board on the Internet, information of the power control device that is not being used by the first user.
5. The server according to claim 4, wherein the power control device includes a plurality of power control devices, and when each of the plurality of power control devices is not being used by the first user, the processor presents, on the bulletin board, information of each of the plurality of power control devices.
6. The server according to claim 5, wherein the processor controls a priority for each of the plurality of power control devices to be lent to the second user, by controlling the order of presentation of the plurality of power control devices on the bulletin board.
7. The server according to claim 6, wherein the processor controls the order of presentation to present, at a different rank on the bulletin board, the power control device included in the plurality of power control devices and installed at a location at which the traffic amount is smaller.
8. The server according to claim 6, wherein the electrically powered vehicle performs the power control in accordance with a supply/demand adjustment request for the power in each of a plurality of regions, and the processor controls the order of presentation to present, at a different rank on the bulletin board, the power control device included in the plurality of power control devices and located in a region in which the supply/demand adjustment request is larger.
9. The server according to claim 5, wherein the plurality of power control devices include a first control device presented on the bulletin board when the first control device is not being used by the

first user, and a second control device not presented on the bulletin board even when the second control device is not being used by the first user, and when supply of the first control device is lower than a demand for the plurality of power control devices, the processor performs control of requesting the first user, who uses the second control device in the normal state, to lend the second control device.

10. The server according to claim 4, wherein the processor presents, on the bulletin board, information of a size of a vehicle body that is able to be parked in a parking area in which the power control device is installed.

11. The server according to claim 1, wherein the processor acquires information of each of an expected traveling route and an expected traveling time period of the electrically powered vehicle, wherein the power control device includes a plurality of power control devices, and when each of the plurality of power control devices is not being used by the first user, the processor selects the power control device to be lent to the second user, based on at least one of the expected traveling route and the expected traveling time period.

12. The server according to claim 1, wherein the processor performs control of requesting the first user of the power control device in a region in which a frequency of lending the power control device to the second user is higher than a predetermined criterion, to permit the power control device to be lent also in a midnight period.

13. A system comprising: a power control device that is configured to perform power control including feeding power to a power system and charging power from the power system; a server that controls the power control device; and an electrically powered vehicle that is configured to perform the power control in response to the electrically powered vehicle being electrically connected to the power control device, wherein the server: acquires information about a status of use of the power control device by a first user who uses the power control device in a normal state, performs, based on the information about the status of use, control for lending the power control device to a second user different from the first user in a time period in which the first user does not use the power control device, and acquires, in response to lapse of a predetermined time, information in real-time of a traffic amount at a location at which the power control device is installed.

14. A method, the method comprising: acquiring information about a status of use of the power control device by a first user who uses the power control device in a normal state; performing, based on the information about the status of use, control for lending the power control device to a second user different from the first user in a time period in which the first user does not use the power control device; controlling the power control device that is configured to perform power control including feeding power from an electrically powered vehicle to a power system or charging power from the power system to the electrically powered vehicle, the electrically powered vehicle being electrically connected to the power control device; and acquiring, in response to lapse of a predetermined time, information in real-time of a traffic amount at a location at which the power control device is installed.

15. The method according to claim 14, further comprising: determining whether a version of a power control program of the electrically powered vehicle coincides with a version of a power control program of the power control device, the determination relating to whether or not there is a connection compatibility between the electrically powered vehicle and the power control device; transmitting the same version of the program as the version of the power control program of the power control device to the electrically powered vehicle in response to determining that the versions are different from each other; and updating the power control program of the electrically powered vehicle in accordance with the transmitted version of the power control program.

16. The server according to claim 1, wherein the processor displays an order of presentation of the power control device in accordance with the acquired information in real-time of the traffic amount.

17. The system according to claim 13, wherein the server displays an order of presentation of the power control device in accordance with the acquired information in real-time of the traffic amount.

18. The method according to claim 14 further comprising displaying an order of presentation of the power control device in accordance with the acquired information in real-time of the traffic amount.
